


objects : In JS, objects are used to group up similar things.



it is an unordered data structure $\Rightarrow$ explained on next page

How to create an object in JS?

let todo = { } ; $\leftarrow$ syntax

object ke andar properties hoti hai.

Key-value pair ko hi hum property kehte hai.

array $\rightarrow$ ordered datastructure



jo pehle likh diya nahi aayega.

Agar array mei 50 daal diya pehle, then "50" 0th position par aayega. Agar 60 likhunga, then 1st position par aayega.

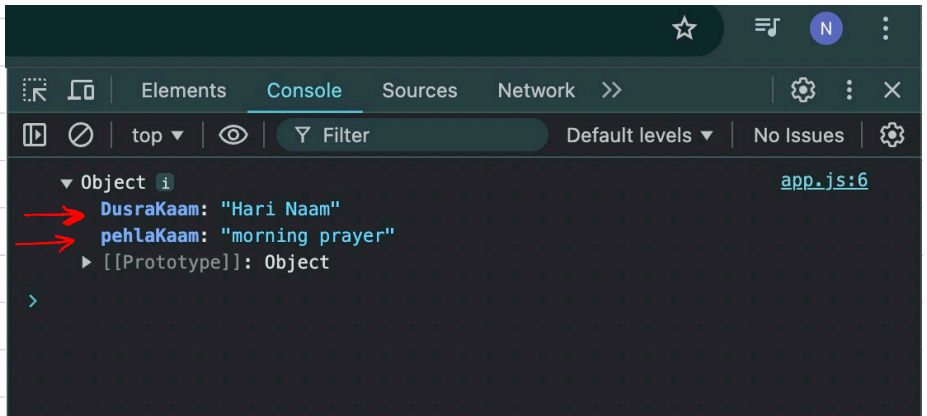
let person = {
 name: "Nitin",
 age: 29
}

property 1
property 2

key
value

! Let's understand why object is an unordered array:-

```
let todo = {  
  pehlaKaam: "morning prayer",  
  DusraKaam: "Hari Naam"  
};  
  
console.log(todo);
```



Humne "pehla kaam" pehle likha tha code mei but o/p mei "pehla kaam" baad mei aaya. Matlab object is unordered data structure.

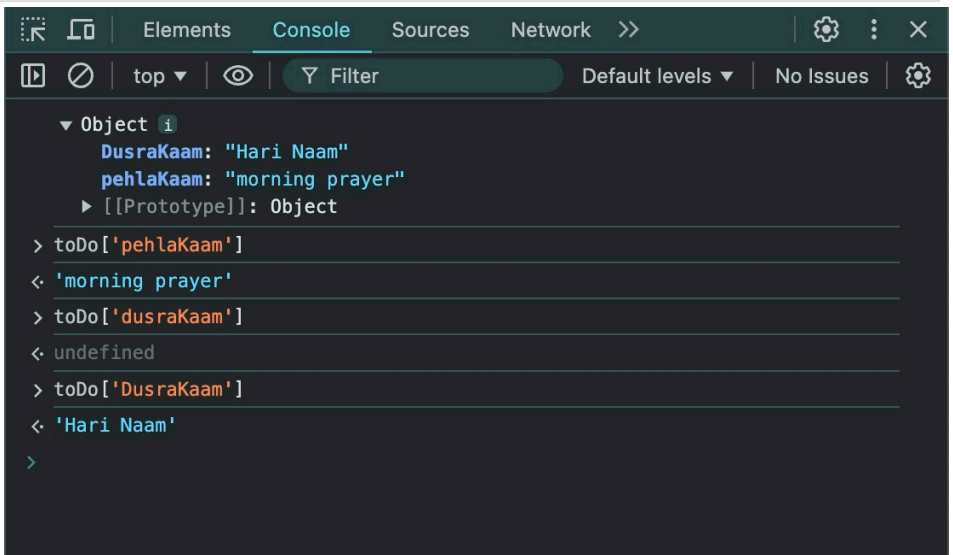
object mai aap indexing use karke access nahi kar sakte. Ex: `todo[1]`, `todo[3]`, etc.

`todo[pehlaKaam]` → ye bhi nahi chalega.

↳ ye hamari "key" ka naam hai

`todo['dusraKaam']` → ye chal jayega

`todo['pehlaKaam']` → this will work



```
Object {
  DusraKaam: "Hari Naam",
  pehlaKaam: "morning prayer",
  [[Prototype]]: Object
}
> todo['pehlaKaam']
< 'morning prayer'
> todo['dusraKaam']
< undefined
> todo['DusraKaam']
< 'Hari Naam'
>
```

single quotes ya double quotes ya backtick use karoge, it will work.

10

```
let todo = {  
  pehlaKaam: "morning prayer",  
  DusraKaam: "Hari Naam",  
  5: true,  
  s5: 'hi there'  
};  
  
console.log(todo);
```

The screenshot shows a web browser's developer console with the 'Console' tab selected. The code being executed is from 'app.js:8'. The console output shows the object `{5: true, pehlaKaam: 'morning prayer', DusraKaam: 'Hari Naam', s5: 'hi there'}`. Below this, several commands are entered in the console, and their results are shown: `> todo[5]` returns `true` (labeled l1), `< true`, `> todo['5']` returns `true` (labeled l2), `< true`, `> todo[s5]` returns `'hi there'` (labeled l3), and `> todo['s5']` returns `'hi there'` (labeled l4). An error message is also visible: 'Uncaught ReferenceError: s5 is not defined at <anonymous>:1:6' (labeled VM271:1).

```
app.js:8  
> {5: true, pehlaKaam: 'morning prayer', DusraKaam: 'Hari Naam', s5: 'hi there'}  
> todo[5] → l1  
< true  
> todo['5'] → l2  
< true  
> todo[s5] → l3  
Uncaught ReferenceError: s5 is not defined at <anonymous>:1:6 VM271:1  
> todo['s5'] → l4  
< 'hi there'  
> |
```

l1, l2 work kar gaya.

l3 → work kyun nahi kiya?

Numbers are super flexible

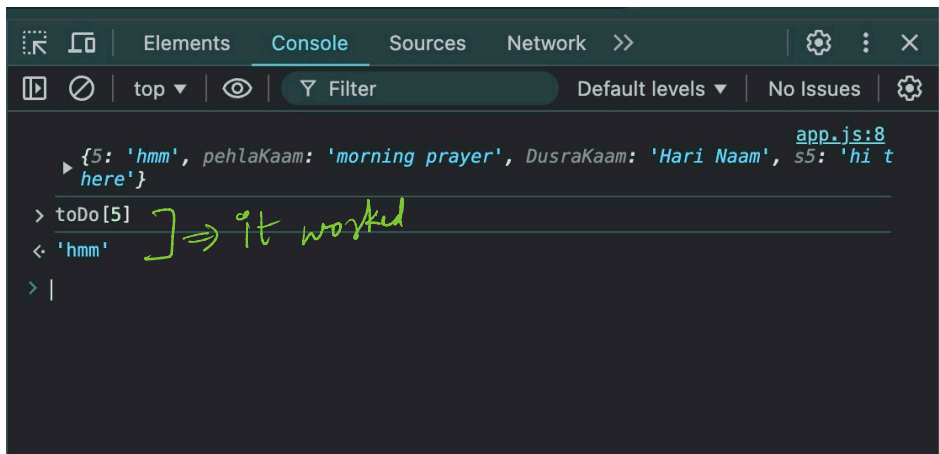
Keys are stored as string in object jab aap number ke alawa kuch bhi dale ho.

jab key is a number (in object), then you can directly access it and vo string ki

tarah bhi stored hota hai.

②

```
let todo = {  
  pehlaKaam: "morning prayer",  
  DusraKaam: "Hari Naam",  
  5: 'hmm',  
  s5: 'hi there'  
};  
  
console.log(todo);
```



```
{5: 'hmm', pehlaKaam: 'morning prayer', DusraKaam: 'Hari Naam', s5: 'hi t  
here'}  
> todo[5]  
< 'hmm'
```

It worked

Important note:-

- ① Don't start with numbers when creating keys in an object. For example I initia as key. You can use this syntax but it is not recommended.
- ② Never use numbers as keys. Ex: 5: 'hmm',
- ③ Keys are stored as string and can be accessed like array but using the same string only. todo ['pehla Kaam']. Numbers is an exception.

④ we always use dot notation if we want to access any value of key.

toDo.pehlaKaam

Dot Notation :-

```
let toDo = {  
  pehlaKaam: "morning prayer",  
  DusraKaam: "Hari Naam",  
  5: 'hmm',  
  s5: 'hi there'  
};  
  
console.log(toDo);  
console.log(toDo.s5);  
console.log(toDo.pehlaKaam);
```

```
app.js:8  
{5: 'hmm', pehlaKaam: 'morning prayer', DusraKaam: 'Hari Naam', s5: 'hi there'}  
hi there  
morning prayer
```

④
let toDo = {
 pehlaKaam: "morning prayer",
 DusraKaam: "Hari Naam",
 5: 'hmm',
 s5: 'hi there'
};

console.log(toDo.pehlaKaam);

toDo.pehlaKaam = 'Siksastakam';

console.log(toDo.pehlaKaam);

→ l1

→ l2

→ l3

```
{5: 'hmm', pehlaKaam: 'morning prayer', DusraKaam: 'Hari Naam', s5: 'hi t  
here'}  
hi there  
morning prayer  
Siksastakam
```

l1 is working as getter. We got the value

l2 is working as setter. We set the value.

todo ['pehlaKaam'] = 'Tulsi arti';

↳ this syntax will work too.

(01:22:00) Break over

object ki key-value pair ko hi object ki property
kehte hai.

Methods :- Functions that are described inside an
object.

```
let person = {  
  naam: 'samarth vohra',  
  place: 'delhi',  
  
  describe: function describe(){  
    console.log('this is describe function');  
  }  
}
```

console.log(person.naam); → l1

console.log(person.describe); → l2

```
samarth vohra  app.js:10
f describe(){  app.js:12
  console.log('this is describe function');
}
```

Humne object ke andar function define kiya.

l2 se → function ki definition console mei print ho gyi.

`console.log ( person.describe() );`

OR

`person.describe() ;`

→ function ko
call karna
padega.

```
let person = {
  naam: 'samarth vohra',
  place: 'delhi',
  describe: function descr(){ → l0
    console.log('this is describe function');
  }
}

console.log(person.naam); → l1
console.log(person.describe); → l2
console.log(person.describe()); → l3
person.describe(); → l4
```

```
samarth vohra → l1 app.js:10
f descr(){ console.log('this is describe function'); } ⇒ l2 app.js:12
}
this is describe function ] ⇒ l3 app.js:6
undefined app.js:14
this is describe function ⇒ l4 app.js:6
```

lo per descr function ko aap access nahi kar sakte.

`person.descr()` → this wont work

aapko `person.describe()` use karna padega.

aap directly `descr()` run karoge object ke bahar, wo bhi nahi chalega. you cannot use object keys & methods outside the object w/o using object name with dot notation.

you can also use this instead of `l0`.

```
describe: function() {
    console.log(...);
}
```

we removed descr name. This too will work fine.

Methods are functions which are described inside the object.

Q guess the output:-

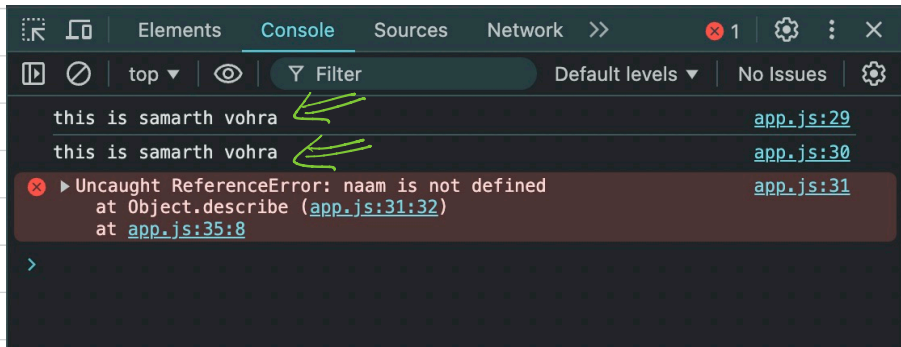
```
let person = {  
  naam: 'samarth vohra',  
  place: 'delhi',  
  
  describe: function(){  
    console.log('this is samarth vohra');  
    console.log(`this is ${person.naam}`);  
    console.log(`this is ${naam}`);  
  }  
}  
  
person.describe();
```

l1
l2
l3

l1 will work

l2 will work

l3 will give error.



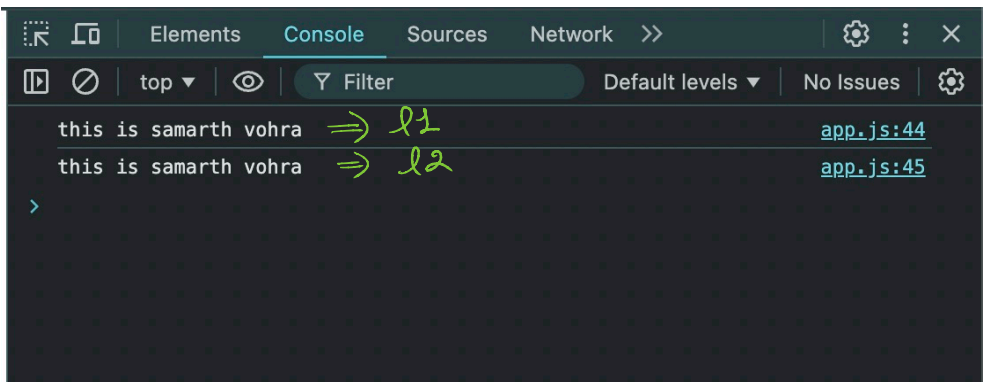
l2 ⇒ Here we used string literal inside function which is inside an object.

There is an alternative to l2:-

```
console.log(`this is ${this.naam}`);
```

Hamne object ka name use nahi kiya, hamne "this" keyword use kiya. This too will work fine. It is a good practice instead of using object name.

```
let person = {  
  naam: 'samarth vohra',  
  place: 'delhi',  
  
  describe: function(){  
    console.log('this is samarth vohra');   
    console.log(`this is ${this.naam}`);   
  }  
}  
  
person.describe();
```



This keyword :- It has many types :-

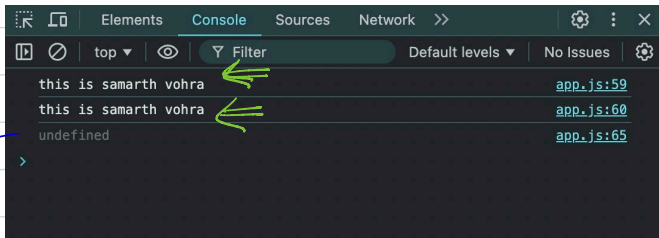
type 1 $\Rightarrow$ this keyword points to the object when referred in the method.

↓
Sir says it is a big lie but the statement is correct in content of the current code.

there are four more types of this keyword.
Baaki ke "this" keyword hum future mei padenge.

① guess the output :-

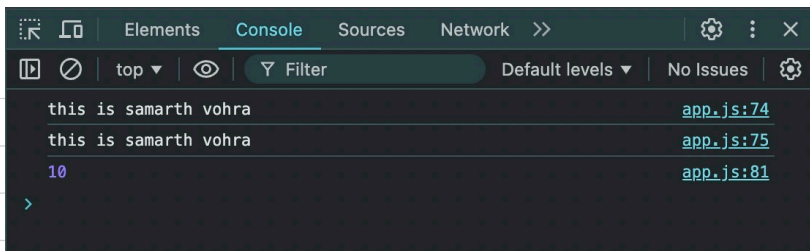
```
let person = {  
  naam: 'samarth vohra',  
  place: 'delhi',  
  
  describe: function(){  
    console.log('this is samarth vohra');  
    console.log(`this is ${this.naam}`);  
  }  
}  
  
let ans = person.describe();  
console.log(ans);
```



→ yaha undefined isliye aaya kyunki function kuch return nahi kar raha. jab function kuch return nahi karata, then undefined console hoga agar aap function ko call karke console.log kar rahe ho.

② guess the output :-

```
let person = {  
  naam: 'samarth vohra',  
  place: 'delhi',  
  
  describe: function(){  
    console.log('this is samarth vohra');  
    console.log(`this is ${this.naam}`);  
    return 10;  
  }  
}  
  
let ans = person.describe();  
console.log(ans);
```

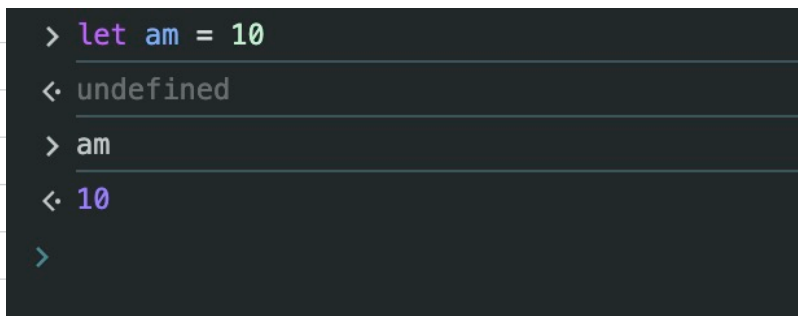


```
this is samarth vohra
this is samarth vohra
10
```

function returned to isliye 10 aaya.

return ki gyi value "ans" mei store ho gyi,
then "ans" ko console log kar diya.

Q Agar mei browser ke console mei let am = 10
likhta hu, then vo undefined return kyu
karta hai.



```
> let am = 10
< undefined
> am
< 10
>
```

Ans Behind the scene, am ko 10 assign karne
ke liye function chal raha hai and vo
function undefined return kar raha hai
kyunki uske function mei return statement
nahi hai.

